Status	Finished
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Question 1

Complete

Which of the following is/are NOT main function(s) of the Data Link Layer?

(Incorrect selections will result in a penalty)

- [ ] a. Framing
- [ ] b. Flow control
- [x] c. Signal encoding
- [ ] d. Error control
- [x] e. Routing
- [x] f. Media access control

Question 2

Complete

Which statement about the Data Link Layer is **NOT** correct?

- a. The Data Link Layer may use framing and control information to support communication between devices on a link.
- b. The Data Link Layer provides communication over a link between neighboring nodes.
- c. The Data Link Layer is responsible for selecting the best path across multiple networks from source to destination.
- d. Error control helps detect and manage errors that may occur during frame transmission.
- e. Flow control helps prevent a sender from transmitting frames faster than the receiver can handle.

Question 3

Complete

Which **TWO** statements **correctly describe** the Data Link Layer sublayers and their use in point-to-point and multipoint communication?
Select two answers. A penalty is applied for each incorrect selection.

- [ ] a. LLC manages logical link functions such as flow and error control between communicating nodes.
- [ ] b. MAC is responsible for selecting the best route between different networks.
- [x] c. LLC is used only in multipoint networks and is unnecessary for point-to-point communication.
- [ ] d. A point-to-point link always requires MAC contention because two devices share the same physical link.
- [ ] e. MAC controls how multiple stations access a shared medium in a multipoint network.
- [x] f. In a multipoint/shared-medium network, only LLC is required because all stations can transmit whenever the medium is idle.

Question 4

Complete

Which statement about data encapsulation and framing at the Data Link Layer is **NOT** correct?

a. Framing removes the original data from the frame and replaces it entirely with control information before transmission.

b. The frame header can contain control information that helps define how communicating devices interpret and manage the frame.

c. The extra control information may be placed in a frame header and trailer surrounding the original data.

d. The original data becomes encapsulated inside a frame before being passed to the Physical Layer for transmission.

e. Framing adds extra control information to the original data so that communicating devices can manage functions such as flow control, error control, and medium access.

Question 5

Complete

Match each description with the appropriate HDLC frame field.

- Used to indicate the beginning and end of an HDLC frame: Flag
- Used to identify the secondary station: Address
- Used for error detection during transmission: FCS
- Used to indicate the frame type and carry control information such as sequence numbers: Control

Question 6

Complete

Which statement about HDLC frame types is **NOT** correct?

a. A **Supervisory frame (S-frame)** is used mainly for flow control and error control.

b. A **Supervisory frame (S-frame)** is primarily used to carry user data between stations.

c. The **Control field** identifies the HDLC frame type and may contain sequence and supervisory information.

d. An **Information frame (I-frame)** carries user data and may also include sequencing and acknowledgment information.

e. An **Unnumbered frame (U-frame)** is used for link management and control functions.

Question 7

Complete

Which statement about HDLC communication modes is **NOT correct**?

a. In ARM, a secondary station may initiate transmission without waiting for a poll from the primary.

b. NRM may be used with point-to-point or multipoint communication involving primary and secondary stations.

c. In ABM, one station must always remain the primary station while the other remains the secondary station.

d. In ABM, both stations operate as combined stations and either side may initiate transmission.

e. In NRM, the primary station controls communication and grants permission to a secondary station before it transmits.

Question 8

Complete

Which statement comparing HDLC with other Data Link/WAN protocols is **NOT correct**?

a. **PPP** uses an HDLC-like frame structure but adds a **Protocol** field to identify the encapsulated upper-layer protocol.

b. **Frame Relay** uses variable-length frames with a Q.922-based address field that includes a DLCI.

c. **Cisco HDLC** is based on HDLC framing and includes a **Type** field to identify the encapsulated protocol.

d. **LAPB** is based on HDLC framing and was used as the Data Link Layer protocol for X.25.

e. **ATM** follows the same HDLC-style variable-length frame format with Flag fields at both ends and an FCS trailer.